

6 Logic

6.1 Sets and Boolean Logic

Definition 6.1.1 A **set** is a collection of distinct objects, called elements of the set. A set can be defined by describing its contents, or listing the elements of the set, in curly brackets.

Describe how we will typically give a set a name, say A , and then we can make statements like $x \in A$. Define $\in$. Define $\emptyset$.

Definition 6.1.2 The **union** of two sets A and B is the set consisting of all elements of A or B . The union of A and B is denoted $A \cup B$.

Definition 6.1.3 The **intersection** of two sets A and B is the set consisting of all elements which belong to both A and B , denoted $A \cap B$.

Example 6.1.4 Find the union and intersection of the two given sets. A is the set of all even numbers. B is the set of whole numbers between 1 and 11 (including 1 and 11).

Definition 6.1.5 A **subset** of a set A is another set B , all of whose elements are in A . Note that B does not have to include *all* the elements of A .

Definition 6.1.6 The **complement** of a set A is the set of elements which do not belong to A (here A must be a subset of some set B). This is denoted A^c or $B \setminus A$.

Example 6.1.7 Is $\{1, 3, 5, 7, 9\}$ a subset of $\{5, 7, 1, 9, 2, 3, -3\}$?

Example 6.1.8 The set $A = \{1, 3, 5, 7, 9\}$ is a subset of $B = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, find $B \setminus A$.

Definition 6.1.9 A **statement** is either true or false.

Definition 6.1.10 Boolean Logic combines multiple statements that are either true or false into an expression that is either true or false.

Mention that three of the most common operations in boolean logic are 'and', 'or', and 'not', and that these correspond to three set operations.

Example 6.1.11 Describe the numbers which meet the condition: "odd number and less than 20 and greater than 0 and (multiple of 3 or multiple of 5)"

Definition 6.1.12 A **conditional** is a compound statement of the form "if p , then q " or "if p , then q , else r ".

Example 6.1.13 In common language, we might say something like the following: "If it's raining, we'll go to the mall. Otherwise, we will go on a hike."

Definition 6.1.14 A **universal quantifier** states that an entire set shares a characteristic.

Definition 6.1.15 A **existential quantifier** states that a set contains at least one element.

Example 6.1.16 Somebody likes baseball.

Example 6.1.17 Suppose your friend says "Everybody cheats on their taxes." What is the minimum amount of evidence you need to prove your friend wrong?

Something interesting happens when you negate a quantifier...